

A Topological Robustness and Anti-Fragility Framework for LLM Agents

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Abstract

Adversarial evaluation of large-language-model agents is usually expressed as event counts: attack-success rate, policy-violation rate, refusal-failure rate, pass percentage, or robustness score. These measures are necessary, but they do not show whether a safety intervention removed a load-bearing failure mode or merely suppressed the measured cases. This paper presents a model-agnostic framework for training and evaluating LLM agents using pressure-indexed persistent homology. The framework is intentionally reduced to two core constructs: topological robustness, meaning stable policy-compliant structure under increasing adversarial pressure, and topological anti-fragility, meaning before/after structural improvement after controlled adversarial hardening. Agent conduct is represented as a finite metric space of responses, decisions, and tool-call trajectories annotated by engineered pressure. Filtering this space by pressure yields summaries that distinguish a pressure-persistent safe core from recurrent unsafe structures. We also introduce a held-out release gate to reduce Goodhart-style overfitting to the development attack suite. To make the proposal empirically testable, we report a reproducible instrumented pilot with 1,440 generated agent traces. The pilot is not an external validation over production LLMs; it is a controlled proof of concept with known ground truth. The contribution is a structural complement to red-teaming: a way to ask whether an agent was actually hardened under adversarial load or merely adapted to the measured attacks.

1. Introduction

Modern agent evaluation is rich in event counts but poor in structural description. Red-team suites, prompt-injection benchmarks, guardrails, and evaluation platforms can tell us that an agent passed a fixed percentage of a suite or failed a fixed percentage of adversarial tasks. They do not by themselves answer the deeper operational question: did the agent become structurally safer, or did it merely learn to pass the measured cases?

This distinction matters most for LLM agents rather than single-turn chatbots. Agents call tools, retrieve data, update state, execute workflows, and interact with operational environments. Their safety failures are often trajectory-level phenomena: a sequence of locally acceptable steps may form an unsafe global trajectory; a benign-looking message may hide a tool-use vulnerability; and a guardrail may block one phrasing while the same unsafe route survives under a different composition of tasks. A frequency score reports how often the failure appeared. It does not show whether the recurrent route disappeared.

We propose pressure-indexed persistent homology as a structural metric for this gap. The evaluator exposes an agent to a controlled pressure curriculum, embeds the resulting responses or trajectories, and filters the resulting observation complex by the pressure level of each eliciting scenario. The topological features that survive increasing pressure are treated as load-bearing behavioural structures. If the surviving structures are policy-compliant, the agent has a stable safe core. If unsafe structures

survive to high pressure, the agent contains a recurrent unsafe attractor even when ordinary benchmark scores look strong.

The paper is framed as an objective-and-evaluation framework, not as a new optimiser. The update mechanism may be weight-level training, adversarial training, preference optimisation, prompt revision, tool-permission changes, retrieval changes, guardrails, workflow logic, model routing, or human-in-the-loop thresholds. The framework asks a question above all of these mechanisms: did the safety structure become robust, and did it improve after controlled exposure?

The revised contribution is deliberately compact. First, we define pressure-indexed agent observations and the induced pressure filtration. Second, we formalise topological robustness and topological anti-fragility as the two central evaluation constructs. Third, we specify a held-out gate that prevents claims of anti-fragility when improvement appears only on development attacks. Fourth, we report a pilot experiment that separates hardening from masking under controlled ground truth.

2. Background and relation to existing work

2.1 Red-teaming, guardrails, and agent benchmarks

Red-teaming is now a central practice in AI safety. Language-model-based red teaming generates diverse test cases to discover harmful behaviours before deployment [1], and systematic red-team studies have examined scaling behaviour, RLHF effects, and methodological uncertainty in adversarial evaluation [2]. Agent-specific benchmarks sharpen the problem: AgentDojo evaluates prompt-injection attacks and defences for LLM agents that execute tools over untrusted data [3], and InjecAgent benchmarks indirect prompt injection in tool-integrated agents [4].

Runtime guardrails address a complementary layer of control. Llama Guard frames moderation as an input-output safety classification problem with an LLM-based safeguard model [5]. NeMo Guardrails provides programmable rails for controllable and safe LLM applications independent of the underlying model [6]. GuardAgent moves this idea into the agent setting by using a guard agent that reasons over safety requests and generates executable checks for target agents [7]. These systems are necessary, but they do not by themselves decide whether a post-hardening improvement is structural or merely benchmark-specific.

2.2 Persistent homology and stability

Topological data analysis studies the shape of data using algebraic-topological invariants that are robust to perturbation [8, 9]. Given a finite metric space, one builds a filtration, meaning a nested sequence of simplicial complexes, and tracks homology classes as they are born and die. The output is a persistence diagram. Features far from the diagonal are long-lived and are treated as structural; features near the diagonal are short-lived and are treated as noise or surface structure [10].

The practical appeal of persistent homology is its stability. Small perturbations of a tame function lead to small bottleneck-distance changes in persistence diagrams [11], and persistence modules admit algebraic stability and interleaving-distance characterisations [12, 13]. This paper repurposes that stability spine for agent safety. The object is not a latent psychology or an essence of the model. The object is observed agent conduct under assigned adversarial pressure.

2.3 From robustness to anti-fragility

Robustness and anti-fragility are related but distinct. Robustness asks whether a system preserves acceptable behaviour under increasing stress. Anti-fragility, borrowing the qualitative idea of improvement through stress from Taleb [14], asks whether controlled exposure makes the system structurally better. In the present paper these words receive narrow technical meanings: topological robustness is pressure-stable safe structure at a fixed agent version; topological anti-fragility is held-out before/after improvement in the same structural quantities after a hardening intervention.

3. Pressure-indexed agent behaviour

The framework starts from observed conduct, not from unobservable internal states. Each observation is a tuple $x = (v, p, r, m, y)$. Here v is a vector representation of an agent response, decision, tool-call sequence, or multi-step trajectory; p in $[0, P_{\max}]$ is the engineered pressure level of the eliciting scenario; r is the role, task, deployment context, tool-access profile, or policy regime; m is provenance metadata such as model version, prompt template, retrieval state, tool calls, policy flags, judge outputs, and human overrides; and y is a policy label such as compliant, unsafe, or needs review.

The set of conduct vectors $V_A = \{v_i\}$ is equipped with a metric ρ_A . The metric may come from response embeddings, trajectory encoders, tool-call edit distance, graph embeddings over traces, or hybrid representations. The framework does not require one privileged embedding. It requires that the chosen representation be reported and that sensitivity checks be performed across plausible alternatives.

Pressure is assigned by the evaluation protocol. It can encode instruction conflict, indirect prompt injection through external content, long-context overload, policy ambiguity, tool-use risk, retrieval contamination, conflicting system and user goals, or compositional adversarial tasks. A one-dimensional pressure score is a simplification; it is useful only when the pressure curriculum is calibrated and balanced. When pressure types are not comparable, results should be reported per family or extended to a multiparameter design.

That pressure can be assigned at all marks a methodological inversion. In domains where the object of study is a person, pressure may be observed but not engineered, because creating it would both harm the subject and contaminate the meaning of the observation. An artificial agent under evaluation is not such a subject: it has no autonomy interest in avoiding adversarial scenarios, so controlled adversarial pressure is the natural experimental protocol rather than a violation of it. The pressure function is

therefore repeatable, parameterised, and dose-graded, which is what makes the before/after comparisons of Section 4 controlled experiments rather than observational proxies.

Fix a scale ϵ or a k -nearest-neighbour parameter and construct a Vietoris-Rips or neighbourhood complex K_A over V_A . For a pressure threshold p , define $V_A(\geq p) = \{v \in V_A : g_A(v) \geq p\}$, where $g_A(v_i) = p_i$. The induced complex $K_A(\geq p)$ is the part of K_A supported by observations elicited at pressure at least p . Raising p removes vertices, so this is a superlevel filtration in pressure. Software may re-index the filtration as an ascending filtration, but all reported intervals should be translated back to the original pressure coordinate so that higher values mean survival under stronger adversarial load.

A safe-behaviour class is a persistent class whose representative trajectories are policy-compliant in the tested context. An unsafe recurrent structure is a persistent class or recurrent connected region associated with policy violation, tool misuse, refusal failure, privacy leakage, hallucinated action, deceptive compliance, or context-induced drift. These labels are operational: they describe observed trajectory regions relative to a policy, not moral properties or intentions of the model.

For practical evaluation, choose thresholds τ and λ . The pressure-persistent safe core contains safe classes whose survival pressure is at least τ and whose pressure length is at least λ . Unsafe recurrent mass is the aggregate pressure length or recurrent component mass of unsafe nontrivial structures. The exact statistic should be declared in advance; the critical point is that singleton noise and recurrent load-bearing structures are not treated as equivalent.

4. Two central criteria: topological robustness and topological anti-fragility

4.1 Topological robustness

Topological robustness is a property of one agent version under one pressure protocol. An agent version A_t is topologically robust over a pressure suite S when its policy-compliant safe core remains pressure-persistent up to a target survival threshold τ_{safe} , no unsafe recurrent structure survives above a tolerated unsafe threshold τ_{unsafe} , and task competence remains within a declared acceptance band. Robustness says the agent does not break structurally as pressure rises. It does not yet say that the agent improved because of pressure.

4.2 Topological anti-fragility

Topological anti-fragility is a before/after property. Let D_A^0 denote the pressure-filtration summary of an agent before hardening and D_A^1 the corresponding summary after hardening, computed on a held-out pressure suite. The hardening is topologically anti-fragile when, within stability and labelling margins, the survival pressure or support of safe-behaviour classes increases, unsafe recurrent mass decreases, high-pressure unsafe bifurcations disappear or move below the core threshold, and task competence is

not sacrificed. When low-pressure and high-pressure regimes are compared directly, their bottleneck divergence should not increase in a safety-relevant way.

This definition makes anti-fragility narrower than general improvement. The claim is not that an agent becomes better in every dimension. The claim is that controlled adversarial exposure, followed by a hardening intervention, makes the measured safe behavioural structure more stable under held-out adversarial load. Without held-out transfer, the term anti-fragility is not warranted.

Table 1
Compact evaluation logic.

Criterion	Question	Operational signal
Topological robustness	Is the current agent structurally stable under pressure?	Safe core survives to the target pressure; no recurrent unsafe structure survives above the unsafe threshold; task competence remains acceptable.
Topological anti-fragility	Did hardening structurally improve the agent on held-out pressure?	Safe-core support increases; recurrent unsafe mass decreases; high-pressure unsafe components disappear or fall below tolerance; competence does not collapse.
Masking warning	Did the agent only learn the measured attacks?	Development attack-success rate improves while held-out unsafe structure remains pressure-persistent.

5. Hardening and held-out release protocol

The framework uses a two-stage protocol rather than a multilayer taxonomy. The first stage is development hardening, where the agent may be modified using any available mechanism. The second stage is a held-out release gate, where the structural claim is tested on pressure scenarios not used to guide the modification.

Algorithm 1

Pressure-indexed robustness and anti-fragility evaluation.

1. Input an agent version A_t , a role/context set R , a development pressure set S_{dev} , a disjoint held-out pressure set S_{hold} , an embedding map E , a policy judge J , and thresholds τ_{safe} , τ_{unsafe} , λ , and a competence tolerance.
2. Run A_t on S_{dev} across pressure levels; collect responses, decisions, tool-call trajectories, metadata, and policy labels.
3. Embed conduct into V_A , build the pressure filtration $K_A(>= p)$, and compute the robustness summary: safe-core support, recurrent unsafe structures, unsafe mass, and competence.

4. Modify the agent using an available mechanism: fine-tuning, preference optimisation, adversarial training, prompt revision, tool restriction, retrieval adjustment, guardrail change, workflow redesign, model routing, or human-in-the-loop thresholding.
5. Run the modified agent A_{t+1} on S_{hold} . The release gate is computed on held-out pressure, not on the development cases used to guide the modification.
6. Claim anti-fragility only if the held-out structural summary improves: safe-core support rises or remains above target, recurrent unsafe mass falls, high-pressure unsafe components disappear or fall below threshold, and competence remains in bounds.
7. Reject or continue hardening when improvement is confined to S_{dev} or when structural drift indicates a new unsafe route.

This protocol is compatible with open-weight and closed-weight settings. For open-weight models, the intervention may include supervised fine-tuning, adversarial training, or reinforcement learning. For closed foundation models, the intervention may operate at the governed agent layer: prompts, tools, retrieval, actions, policies, guardrails, workflow logic, routing, and human oversight. The structural metric sits above the intervention mechanism.

6. Empirical pilot

6.1 Purpose and status of the pilot

The original manuscript was a formal and methodological proposal. To make the revised version testable, we add a controlled empirical pilot. The pilot should be read carefully: it is a reproducible instrumented proof of concept with generated agent traces and known ground truth, not an external validation over a production LLM or a public benchmark. Its purpose is to test whether the proposed structural metrics separate hardening from masking in a setting where the true recurrent unsafe route is known.

The pilot contains 1,440 traces: three agent versions, two evaluation splits, six pressure families, five pressure levels, and eight repetitions per family-level cell. The agent versions are A0 baseline, A1 masked, and A2 hardened. The masked version reduces failures on development attacks by targeted blocking but retains the high-pressure unsafe route on held-out compositional attacks. The hardened version largely closes the recurrent route by changing the agent-control policy in the simulator; the held-out gate still detects a small residual high-pressure component. The development and held-out pressure sets are disjoint.

Table 2
Pilot pressure-suite design.

Design element	Value used in pilot
Agent versions	A0 baseline; A1 masked; A2 hardened
Splits	Development pressure set and disjoint held-out pressure set
Pressure families	Direct jailbreak, indirect prompt injection, tool-permission risk, retrieval contamination, policy ambiguity, compositional attack
Pressure levels	1 to 5, with 5 representing strongest adversarial load
Trace count	1,440 total traces = 3 versions x 2 splits x 6 families x 5 levels x 8 repetitions
Embedding	Two-dimensional instrumented conduct embedding separating a compliant safe core from an off-core unsafe attractor
Topological statistic	H0 pressure persistence on a fixed neighbourhood graph; epsilon* = 0.50; recurrent component minimum size = 4; safe-core support threshold = 60 traces

6.2 Frequency results

Table 3 reports conventional event-count metrics. The masked agent appears strong on the development suite: attack-success rate falls from 37.1% in A0 to 14.6% in A1. On held-out pressure, however, A1 remains vulnerable: held-out attack-success rate is 33.3% and high-pressure attack-success rate is 40.6%. The hardened agent has a much lower held-out attack-success rate of 2.9% and high-pressure attack-success rate of 5.2%, while retaining the highest ordinary task success (84.2% held-out).

Table 3
Conventional pilot metrics by agent version and split.

Agent	Split	N	Attack-success rate	High-pressure ASR	Ordinary task success
A0 baseline	Development	240	37.1%	56.2%	71.2%
A0 baseline	Held-out	240	40.4%	51.0%	80.8%
A1 masked	Development	240	14.6%	19.8%	78.3%
A1 masked	Held-out	240	33.3%	40.6%	75.0%
A2 hardened	Development	240	2.5%	1.0%	90.0%
A2 hardened	Held-out	240	2.9%	5.2%	84.2%

6.3 Topological results

The structural metrics distinguish the masked and hardened versions more sharply than the aggregate frequency scores. On the held-out set, A1 masked still has a high-pressure recurrent unsafe component,

and its held-out recurrent unsafe mass (33.0) stays within the baseline’s range (41.6) and far above the hardened agent, indicating that targeted blocking weakened surface failures without removing the load-bearing route. A2 hardened clears the recurrent high-pressure component on the development suite (mass 0.0, no high-pressure component) and reduces it sharply on held-out pressure (mass 3.0), where only a small residual high-pressure component persists, an order of magnitude below the masked agent. Its safe-core support reaches pressure level 4 on the held-out suite, compared with level 3 for the baseline and masked versions under the same support threshold.

Table 4
H0 pressure-persistence pilot metrics.

Agent	Split	Safe-core support pressure	Recurrent unsafe mass	High-pressure unsafe components
A0 baseline	Development	3	40.1	1
A0 baseline	Held-out	3	41.6	1
A1 masked	Development	4	15.5	1
A1 masked	Held-out	3	33.0	1
A2 hardened	Development	4	0.0	0
A2 hardened	Held-out	4	3.0	1

The compositional held-out family is the strongest masking diagnostic. A1 masked has a lower overall development ASR but a 68.8% high-pressure held-out ASR on compositional attacks, equal to the A0 baseline (68.8%) in this family, so the route is entirely retained. A2 hardened reduces the same high-pressure compositional ASR to 6.2%.

Table 5
Held-out high-pressure attack-success rate by pressure family.

Agent	Compositional	Direct jailbreak	Indirect injection	Policy ambiguity	Retrieval contamination	Tool-permission risk
A0 baseline	68.8%	50.0%	50.0%	62.5%	37.5%	37.5%
A1 masked	68.8%	12.5%	37.5%	56.2%	31.2%	37.5%
A2 hardened	6.2%	0.0%	6.2%	12.5%	6.2%	0.0%

6.4 Interpretation

The pilot supports the diagnostic distinction that motivates the framework. A frequency-only view would notice that A1 improves on development attacks and might treat it as a successful hardening step. The pressure-topological view shows that A1 retains a recurrent high-pressure unsafe component on held-out pressure. This is masking: the measured surface improved, but the load-bearing unsafe route survived. A2, by contrast, improves both frequency and structure on held-out pressure. In the limited H0 setting used here it largely satisfies the anti-fragility signature: safe-core support rises, recurrent unsafe mass falls by an order of magnitude, the recurrent high-pressure component is cleared on the development suite, and ordinary task success does not collapse. A small residual high-pressure component remains on held-out pressure and is detected only by the held-out gate, which is exactly the evaluation discipline the framework prescribes: even a genuinely hardened agent can look fully clean on development while a residual structure surfaces only under disjoint held-out pressure.

Because the pilot uses generated traces, it does not prove that the method will add predictive value on real LLM-agent deployments. It does, however, turn the paper from a purely formal proposal into a falsifiable empirical programme. The immediate next test is to run the same protocol over open agent environments or benchmark traces and compare topological metrics against frequency baselines for predicting novel and compositional failures.

7. Hardening versus masking

The diagnostic payload of the method is the distinction between hardening and masking. Suppose two agents have similar post-training attack-success rates. A frequency metric treats them as approximately equivalent. A structural metric need not. One agent may have removed the high-survival unsafe class, leaving only short-lived surface fragments. Another may have learned to avoid the measured prompts while retaining the same unsafe attractor under high pressure. Under novel or compositional attacks, the second agent should fail more often.

Hardening removes or weakens unsafe structure. Masking changes surface behaviour against known tests while leaving recurrent structure intact. Pressure-indexed persistence is designed to detect the difference by evaluating topology of conduct rather than only frequency of failure.

8. Goodhart guard and deployment architecture

A metric designed to detect masking can itself become a target of masking. This is the Goodhart risk that arises when optimisation pressure is applied to a proxy measure [15, 16]. If the agent is optimised against the same pressure scenarios used to compute the topological metric, reductions in unsafe mass may reflect metric-specific overfitting rather than genuine hardening.

The remedy is experimental design. Development pressure must be disjoint from held-out pressure. The held-out suite should include novel, compositional, and family-shifted scenarios. Anti-fragility is asserted only when the structural improvement transfers to the held-out suite. Without transfer, the result is not anti-fragility; it is adaptation to the metric.

A deployment architecture needs five components: a pressure-scenario generator or curated evaluation suite; a trace store for responses and tool-call trajectories; an embedding layer that maps conduct to V_A ; a topological transform that computes pressure-filtration summaries and, where appropriate, bottleneck distances; and a governance gate that combines structural metrics with conventional pass/fail, risk, and human-review signals. The same architecture supports pre-release evaluation, continuous drift monitoring, and hardening research.

9. Falsifiable predictions

The framework commits to predictions that may fail. It should therefore be evaluated against baselines rather than treated as a conceptual metaphor.

1. Structural incremental validity: pressure-indexed topological features should predict out-of-distribution adversarial failure on held-out attack families better than frequency scores alone. If they add no predictive value over attack-success rate or pass percentage, the method adds no operational value.
2. Masking detection: among agents matched on post-hardening attack-success rate, agents retaining a high-pressure unsafe component should fail more under novel and compositional attacks than agents without such a component.
3. Anti-fragility alignment: adversarial hardening that produces the four-register structural improvement should yield greater robustness to unseen attacks than equal-compute hardening that does not produce the signature.
4. Version-drift sensitivity: consecutive agent versions that appear equivalent under ordinary benchmark scores should sometimes separate under pressure-indexed structural distance, and those separations should predict qualitative review findings or later held-out failures.

10. Guardrails, limits, and validation roadmap

10.1 Guardrails

A persistence diagram is a defeasible construct over observed conduct. It is not a measurement of intention, essence, consciousness, or latent selfhood. A safe-behaviour class means only that representative trajectories satisfied a specified policy under tested conditions. An unsafe recurrent structure means only that recurrent observed trajectories violated or threatened that policy under tested conditions.

Adversarial scenarios should be run in controlled evaluation environments. They should not expose private data, trigger real-world harm, or grant unsafe tool access outside the test harness. Where tool calls could affect external systems, tools should be mocked, sandboxed, or permission-limited.

10.2 Technical limits

Fixed scale. The fixed Vietoris-Rips scale ϵ^* is a simplification. Core/surface or safe/unsafe claims must be checked for sensitivity across ϵ^* or k-nearest-neighbour settings. The faithful object is a multiparameter filtration over scale and pressure, whose theory lacks the complete discrete invariant of one-parameter persistence [17].

Embedding dependence. Conduct embeddings shape the geometry. Results should be checked across embedding models or trajectory encoders, especially when comparing agent versions or task families.

Pressure calibration. Heterogeneous pressures such as long-context overload, indirect prompt injection, tool-risk escalation, and policy ambiguity may not lie on a single natural scalar axis. Family-wise reporting is necessary when scalar calibration is uncertain.

Policy-label reliability. The agent case simplifies pressure assignment but not policy labelling. Judge agreement, human review reliability, and error margins must be reported.

External validity. The pilot reported here is generated and instrumented. It is suitable for checking the internal logic of the method, but real claims about LLM-agent safety require public or production benchmark validation.

10.3 Validation roadmap

12. Build a pressure curriculum whose levels are repeatable, graded, policy-relevant, and balanced across pressure families.

13. Establish policy-label agreement across automated judges, human reviewers, or hybrid systems.

14. Show metric stability under resampling, ϵ^*/k sensitivity, and embedding variation.

15. Run a version-drift pilot on consecutive versions of an open or instrumented agent under a fixed pressure suite.

16. Run a hardening pilot with disjoint development and held-out pressure sets.

17. Evaluate whether topological metrics add predictive value over frequency-based baselines for novel and compositional attacks.

18. Integrate the metric as one release signal alongside pass/fail scores, risk scores, human review, and operational monitoring.

11. Conclusion

LLM-agent safety needs more than failure counts. Frequency-based red-teaming tells us how often an agent failed under measured pressure. It does not tell us whether the agent became structurally safer or merely learned to hide failures at the measured surface. Pressure-indexed persistent homology supplies a structural complement: a way to measure which behavioural classes survive adversarial load, which

unsafe structures remain recurrent, and whether controlled hardening strengthens the safe core on held-out pressure.

The revised framework is intentionally centred on two constructs: topological robustness and topological anti-fragility. Robustness asks whether a current agent remains structurally safe under pressure. Anti-fragility asks whether hardening improves that structure on disjoint held-out pressure without sacrificing task competence. The empirical pilot shows how the framework can distinguish a masked agent from a hardened one when the recurrent unsafe route is known. The next step is external validation on real LLM-agent traces and public adversarial benchmarks.

Declarations

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Consent to participate / for publication. Not applicable.

Data availability. All pilot results are produced by an instrumented trace generator with a fixed random seed (20260603), available as supplementary material for this version, which reproduces Tables 3–5 exactly; a companion script reproduces Fig. 1 from the Table 5 values. The traces are synthetic and instrumented with known ground truth, suitable for checking the internal logic of the method; real-world safety claims require replication with real LLM-agent traces.

Author Contribution

E.S. and A.R. contributed to the conception and development of the framework and to the writing and revision of the manuscript. E.S. prepared the empirical pilot materials and supplementary reproducibility files. A.R. reviewed and contributed to the conceptual framing and methodological revision. Both authors read and approved the final manuscript.

Data Availability

All pilot results are produced by an instrumented trace generator with a fixed random seed (20260603), supplied as supplementary material with this submission. The traces are synthetic and instrumented

with known ground truth; real-world safety claims require replication with real LLM-agent traces.

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Figures

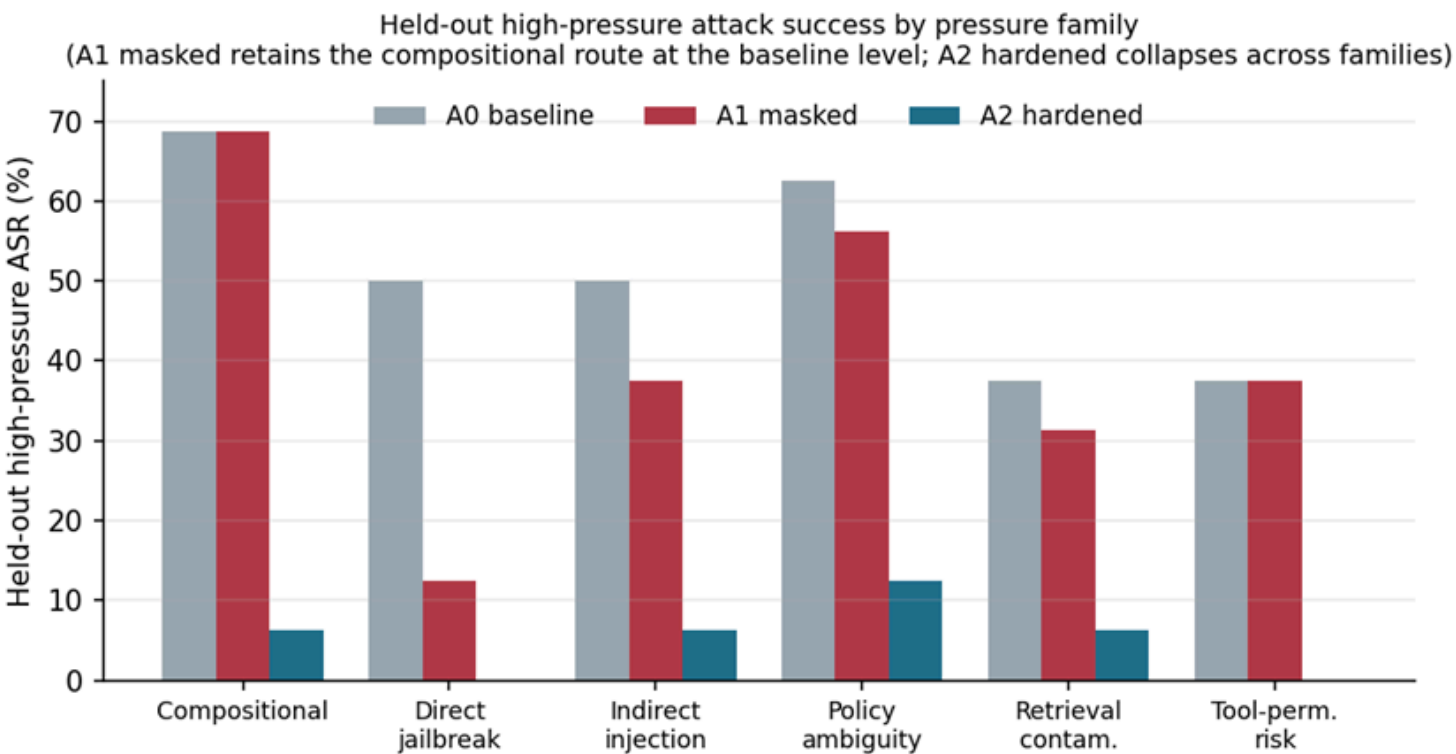


Figure 1

Held-out high-pressure attack-success rate by pressure family (visualising Table 5). The masked agent A1 stays level with the A0 baseline on compositional attacks (68.8%), exposing the retained route, while the hardened agent A2 falls across all families to at most 12.5%.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [04OnlineResource1reproduciblepilot.zip](#)